

How BMS Handles Your Money & How the Four Reports Are Built

A plain-language guide to every place money moves in and out — and exactly where the Income Statement, Balance Sheet, Cash Flow Statement and Trial Balance get each number.

BJP Technologies Co. Ltd · Business Management System (BMS)

Written so anyone can read it, enter test data, and confirm the numbers are real.

What this document is for

If you have ever asked *“When I receive or pay money, what does the system actually record?”* or *“Where does this figure on the report come from — is it real or made up?”* — this guide answers both, end to end.

It is deliberately practical. After each section there is a **“Try it yourself”** box: enter a small amount of test data, then watch the exact numbers it should produce. If reality matches the guide, the books are telling the truth.

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1. The one big idea (read this first)

Every shilling that moves is written into one shared book — the General Ledger — using double-entry. All four reports are just different views of that same one book. Nothing is typed into a report by hand.

In accounting there is a golden rule: **nothing happens on only one side**. If money lands in the bank, it had to come *from* somewhere (a customer paying, a sale, an owner's deposit). So every event is recorded as at least two lines that are equal and opposite — a **Debit**

and a matching **Credit**. They must always sum to the same total. That is “double-entry”.

In BMS this shared book is two database tables working together:

journal_entries

One row per event (the date, description, who did it, which project, and a **status**).

journal_entry_items

The lines of that event — each one is a **debit** or **credit** on a specific account, with an amount.

Two rules make the whole system trustworthy:

- **One door in.** Every money event is posted through a single helper, `postLedgerEntry()` . It refuses to save anything unless the debits equal the credits. So the book can never go crooked one event at a time.
- **Only “posted” counts.** An entry has a status: `draft` , `posted` , `void` or `reversed` . **Every report reads only `posted` entries.** Drafts and voided items are ignored — they are work-in-progress, not the truth.

Why this matters (a real example). Until recently the Trial Balance accidentally added up some *un-posted* lines, which made it look ~627 million out of balance. The fix was to count only posted lines. After the fix the books balance to the cent (Debits = Credits = exactly the same number). This is the difference between a report that *looks* official and one that is actually *true*.

Each account knows what kind of thing it is, from the `account_types` table. That single fact decides which report — and which section of it — the account belongs to:

ACCOUNT CATEGORY	NATURAL SIDE	APPEARS ON	EVERYDAY MEANING
asset	Debit	Balance Sheet	Things you own / are owed (bank, stock, customers' debts)
liability	Credit	Balance Sheet	Things you owe (suppliers, taxes, customer deposits)
equity	Credit	Balance Sheet	The owners' stake
revenue	Credit	Income Statement	Ordinary sales income
other_income	Credit	Income Statement	Income that isn't your main business (interest, gains)
cogs	Debit	Income Statement	Cost of the goods you sold

ACCOUNT CATEGORY	NATURAL SIDE	APPEARS ON	EVERYDAY MEANING
expense	Debit	Income Statement	Running costs (rent, salaries, fuel)
finance_cost	Debit	Income Statement	Bank charges & financing costs

2. Debit & Credit in everyday words

You don't need to be an accountant to read the reports, but two words help:

Debit = value coming IN to an account

Cash account goes up → debit it. A cost happens → debit the expense. (Debit-natural accounts: assets, expenses.)

Credit = value going OUT of an account

Cash account goes down → credit it. A sale earns income → credit the revenue. (Credit-natural accounts: liabilities, equity, income.)

That's the whole trick. Below, every money event is shown as its Debit/Credit pair so you can see the two sides with your own eyes.

3. Money coming IN — every door

There are several ways money (or a promise of money) arrives. Each one posts its own balanced entry into the one ledger. Here is every inflow door, in plain terms, with the exact double-entry it writes.

WHEN THIS HAPPENS...	...THE SYSTEM RECORDS	STATE
You approve a sales invoice the moment a sale is earned, before cash arrives	Dr Accounts Receivable (full invoice incl. VAT) Cr Sales Revenue (the goods/service value) Cr Output VAT Payable (18% you owe TRA)	posts

WHEN THIS HAPPENS...	...THE SYSTEM RECORDS	STATE
A customer pays an invoice single payment, or a receipt voucher covering many	<pre>Dr Bank / Cash (money received) Cr Accounts Receivable (their debt clears) // if the customer withheld tax: Dr Bank (net) + Dr WHT Receivable / Cr AR (gross)</pre>	posts
Other income received e.g. interest, a small non-sales receipt	<pre>Dr Bank / Cash Cr Income account (+ Cr Output VAT if taxable)</pre>	posts
A customer pays in advance / deposit money before any invoice exists	<pre>Dr Bank / Cash Cr Client Deposits (a liability – you owe them goods) // later, when applied to an invoice: Dr Client Deposits / Cr Accounts Receivable</pre>	posts
A project milestone is certified (IPC) construction-style interim payment certificate	<pre>Dr Accounts Receivable Cr Contract Revenue (the certified amount)</pre>	posts
A point-of-sale (POS) sale retail counter sale	<pre>Dr Cash / Cr Sales / Cr Output VAT Dr Cost of Goods Sold / Cr Inventory</pre>	being wired

Tanzania reality (built into these doors). VAT is **18%** and a **fiscalised EFD receipt** must be issued when payment is received. Most customers pay by **mobile money** (M-Pesa, Mixx, Airtel Money), bank transfer, or cash. Where a customer or a government body **withholds tax**, the withheld part is recorded separately (WHT Receivable) instead of cash. These rules come from TRA and the Finance Act — see [references](#).

Try it yourself — prove a sale and a payment are real

1. Create a sales invoice for a customer: 1 item at **100,000**, VAT 18% → grand total **118,000**. **Approve** it.
2. Open **Reports** → **Income Statement**. **Sales Revenue** should rise by **100,000** (the VAT is *not* income — it's money you hold for TRA).
3. Open **Reports** → **Balance Sheet**. **Accounts Receivable** rises by **118,000** and **Output VAT Payable** (a liability) rises by **18,000**.
4. Now **record a payment** of 118,000 into your bank account.
5. Re-open the Balance Sheet: Receivable drops back by 118,000, **Bank rises by 118,000**. Open the **Cash Flow** — that 118,000 now shows as cash received from operations.

If every one of those moved exactly as written, the money-in path is sound.

4. Money going OUT — every door

Same principle in reverse. Each outflow posts a balanced entry. Most expenses post in **two stages** — first the cost is *recognised* (you owe it), then later it is *paid* (cash leaves). This is “accrual accounting”, and it is what makes the Income Statement honest about costs even before cash moves.

WHEN THIS HAPPENS...	...THE SYSTEM RECORDS	STATE
An expense or payment voucher is approved the cost is incurred	Dr Expense account Cr Accrued Expenses (you now owe it)	posts
...and then it is paid	Dr Accrued Expenses (the debt clears) Cr Bank / Cash (money leaves)	posts
Goods received from a supplier (GRN)	Dr Inventory Cr Accounts Payable (you owe the supplier)	posts
You pay a supplier	Dr Accounts Payable Cr Bank / Cash	posts

WHEN THIS HAPPENS...	...THE SYSTEM RECORDS	STATE
Payroll is run salaries earned, taxes withheld	Dr Salaries Expense Cr PAYE Payable / Cr NSSF / Cr SDL / Cr Salaries Payable // pay day: Dr Salaries Payable / Cr Bank	posts
Statutory remittance PAYE / NSSF / SDL / VAT / WHT sent to authorities	Dr the relevant Payable Cr Bank	posts
Petty cash spend / top-up	spend: Dr Expense / Cr Petty Cash top-up: Dr Petty Cash / Cr Bank	posts
Bank transfer between own accounts	Dr Bank-B / Cr Bank-A (+ Dr Bank Charges if a fee)	posts
Buy a fixed asset · run depreciation · dispose	buy: Dr Fixed Asset / Cr Bank or Payable depreciate: Dr Depreciation Expense / Cr Accumulated Depreciation dispose: Dr Cash + Dr Accum. Dep. / Cr Asset (+ gain/loss)	posts

Tanzania reality (built into these doors). An expense needs a valid **EFD receipt** to reclaim input VAT. Payroll carries the full statutory set — **PAYE** (progressive, first 270,000/month tax-free), **NSSF 20%** (10% employer + 10% employee), **SDL 3.5%** (employers with ≥10 staff) and **WCF 0.5%** — each remitted to its authority by the deadline. Supplier payments may carry VAT-withholding (3% goods / 6% services) and 2% government WHT.

Try it yourself — prove an expense touches the right places

1. Create an expense, e.g. **Fuel 50,000**, and **approve** it (don't pay yet).
2. **Income Statement** → Expenses rise by 50,000. **Balance Sheet** → **Accrued Expenses** (a liability) rises by 50,000. Cash has *not* moved yet — correct.
3. Now **mark it paid** from the bank.
4. **Balance Sheet** → Accrued Expenses drops 50,000, **Bank drops 50,000**. **Cash Flow** → shows 50,000 cash out under operating. Income Statement is unchanged (the cost was already counted at step 1). That two-step behaviour is accrual accounting working.

5. The four reports — subsection by subsection

All four read the **same one ledger** (posted entries only) and group it by the account categories in the table above. They differ only in *which accounts* they show and *how* they arrange them. Here is exactly where each subsection's number comes from.

One engine, four views. The calculations live in one file, `core/financial_reports.php`, with one function per report: `glProfitLoss()` (Income Statement), `glBalanceSheet()`, `glCashFlow()`, and `glTrialBalance()`. Because they share the same source, they cannot disagree — and a built-in guard, `assertLedgerBalanced()`, can prove at any moment that Debits = Credits *and* Assets = Liabilities + Equity.

5a. Income Statement (also called Profit & Loss)

Question it answers: “Over this period, did we make a profit?” It shows income earned and costs incurred *between two dates* — so it uses the **flow** in the window, never opening balances.

SUBSECTION	WHERE THE NUMBER COMES FROM	HOW IT'S CALCULATED
Revenue	All accounts of category <code>revenue</code>	Credits – Debits in the period (income posted by invoice approval / IPC / POS)
Other Income / Gains	category <code>other_income</code>	Credits – Debits — kept separate from sales, per IFRS (revenue vs gains)
Cost of Goods Sold	category <code>cogs</code>	Debits – Credits — the cost of items sold
Gross Profit	<i>derived</i>	Revenue – Cost of Goods Sold

SUBSECTION	WHERE THE NUMBER COMES FROM	HOW IT'S CALCULATED
Operating Expenses	category <code>expense</code>	Debits – Credits (rent, salaries, fuel...)
Finance Costs	category <code>finance_cost</code>	Debits – Credits (bank charges)
Net Profit	<i>derived</i>	(Revenue + Other Income) – (COGS + Expenses + Finance Costs)

Live page: **Reports** → **Income Statement** → `get_income_statement.php` → `glProfitLoss()`. Drill-down on any line lists the very journal entries behind it.

5b. Balance Sheet (Statement of Financial Position)

Question it answers: “On this date, what do we own, what do we owe, and what's left for the owners?” It shows **cumulative** balances up to the chosen date (everything since inception).

SUBSECTION	WHERE THE NUMBER COMES FROM	HOW IT'S CALCULATED
Assets bank, receivables, stock, fixed assets	Every <code>asset</code> account	Debits – Credits, cumulative to the date
Liabilities suppliers, taxes owed, deposits held	Every <code>liability</code> account	Credits – Debits, cumulative
Equity (paid-in)	Every <code>equity</code> account	Credits – Debits, cumulative
Retained Earnings	<i>derived from the Income Statement side</i>	All income – all costs, from inception to the date — folded into equity automatically
Balanced?	<i>the integrity test</i>	Assets must equal Liabilities + Equity. If not, the page says so plainly.

Why it always balances (when the data is clean). Because every entry has Debits = Credits, it's a mathematical fact that **Assets = Liabilities + Equity + (Income – Costs)**. The Balance Sheet folds “Income – Costs” into equity as **Retained Earnings**, so the two sides match by construction. If they ever *don't*, the cause is bad opening data or an unclassified account — and the page surfaces it instead of hiding it.

Live page: **Reports** → **Balance Sheet** → reads the posted ledger the same way `glBalanceSheet()` does.

5c. Cash Flow Statement

Question it answers: “Where did actual cash come from and go, this period?” Profit is not cash (you can be profitable but cash-poor). This statement tracks only the bank & cash accounts.

SUBSECTION	WHERE THE NUMBER COMES FROM	MEANING
Operating activities	Cash entries whose other side is income, an expense, a receivable/payable, stock	Cash from the day-to-day business
Investing activities	Cash entries whose other side is a fixed asset (codes 1-3xxx / non-current)	Buying or selling long-term assets
Financing activities	Cash entries whose other side is equity	Owner capital in, drawings out
Net change in cash	<i>the proof</i>	Operating + Investing + Financing — must equal the bank balance movement
Opening + change = Closing	cash accounts at start and end	Ties back to the Balance Sheet's cash line

How it's built (the clever bit). The system looks at every entry that touched a cash/bank account, then reads the *other* side of that entry to decide the category. Because debits always equal credits, the three sections are guaranteed to add back up to the real change in the bank balance. So this statement reconciles by construction — it can't drift away from reality like a hand-built one can.

Live page: **Reports** → **Cash Flow** → `cash_flow_gl.php` → `get_cash_flow.php` → `glCashFlow()` . Direct method (IAS 7).

5d. Trial Balance

Question it answers: “Is the whole ledger internally consistent — do all debits equal all credits?” It is the accountant's pre-flight check before trusting the other three. It lists **every** account with a balance and proves the totals match.

SUBSECTION	WHERE THE NUMBER COMES FROM	HOW IT'S CALCULATED
Each account row	Every account with any opening balance or posted activity	Its net balance, placed in the Debit or Credit column by which side it lands on

SUBSECTION	WHERE THE NUMBER COMES FROM	HOW IT'S CALCULATED
Section subtotals	grouped by category	Assets, Liabilities, Equity, Revenue, Other Income, COGS, Expenses, Finance Costs
Grand Total Debit / Credit	<i>the proof</i>	The two columns must be equal. If not → a red “does not balance” banner with the difference.
Net Profit (derived)	the income & cost sections	(Revenue + Other Income) – (COGS + Expenses + Finance Costs) — the same figure the Income Statement shows

The bug that was just fixed (so you can trust this page). The Trial Balance used to accidentally include un-posted (draft) lines in its totals, showing a phantom **≈627,000,000** imbalance. It now counts **posted entries only**. Verified on the live data, the two columns are now **exactly equal** (Debit = Credit = 15,474,270,147.68; difference 0.00).

Live page: **Reports** → **Trial Balance** → `app/constant/reports/trial_balance.php`, mirroring `glTrialBalance()`.

Try it yourself — confirm the books are honest

1. Open **Reports** → **Trial Balance**. The **Total Debits** and **Total Credits** cards should show the *same* number, and “Balance Integrity” should read **BALANCED**.
2. Note the **Net Profit** figure at the foot. Open the **Income Statement** for the year — the Net Profit should match.
3. Open the **Balance Sheet**. **Retained Earnings** should equal that same all-time Net Profit, and **Assets = Liabilities + Equity**.

When all three agree, you've proven the four reports are different windows onto one true book.

6. The big picture — where every report gets its data

Money events on the left all pass through the **one door** into the **one ledger**. The four reports on the right are simply different readings of that same posted ledger.

 Money IN
  Money OUT
  The one ledger
  Reports

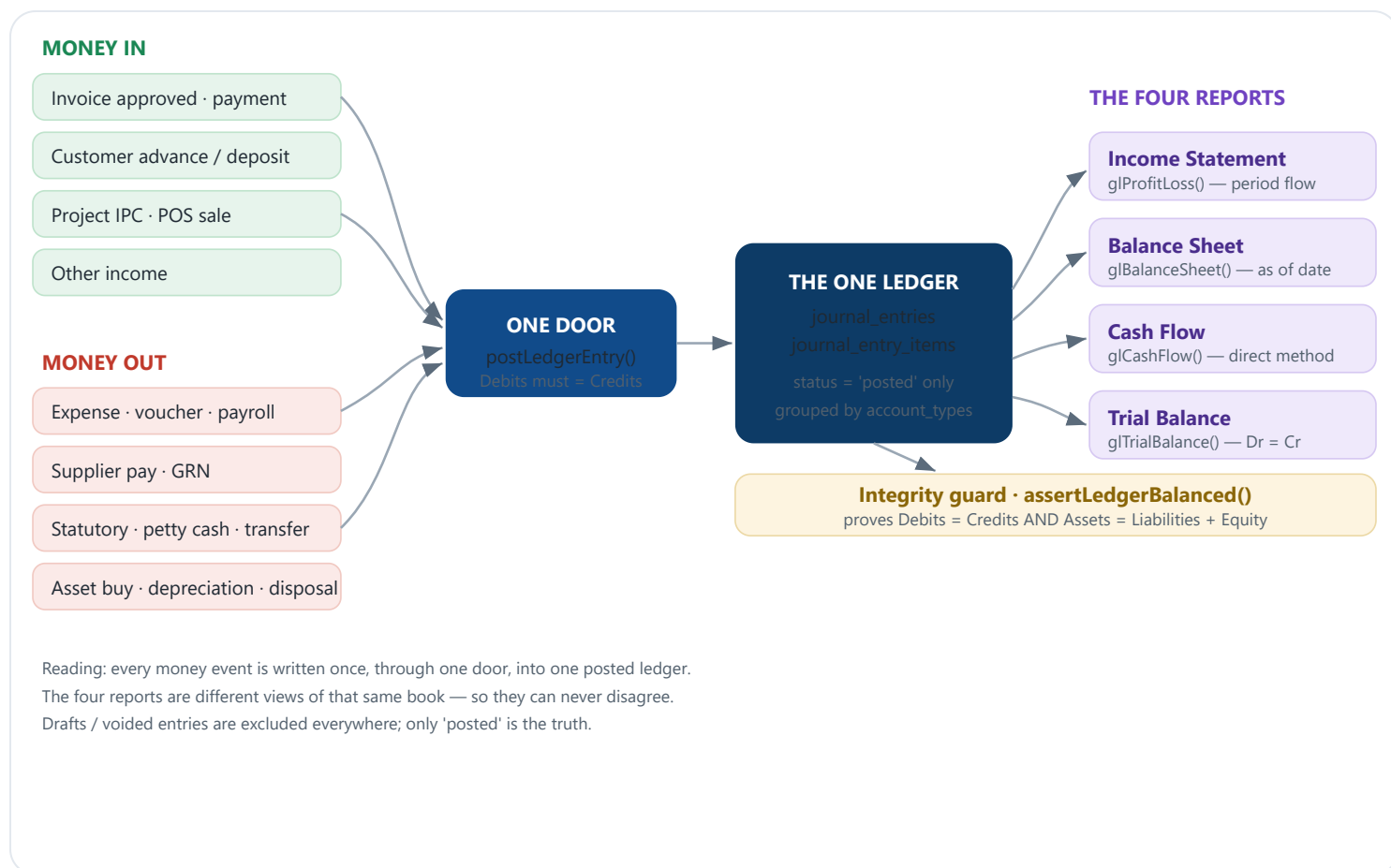


Figure 1 — From a money event to the four reports. One door in, one ledger, four views out.

7. Standards & references used

The design follows recognised accounting standards and Tanzanian rules, not invention:

TOPIC	WHAT WE FOLLOWED	SOURCE
Double-entry, one ledger, Trial Balance → financial statements	The professional pattern (one posted ledger feeding all statements), mirrored from open-source double-entry accounting (AccountGo / WorkDo DoubleEntry).	money.md F1/F3
Revenue recognised when earned; revenue vs other gains separated	IFRS 15 / IFRS for SMEs §23 (revenue) and the revenue-vs-gains distinction.	ifrs.org
Balance Sheet & Income Statement presentation	IAS 1 / IFRS for SMEs.	NBAA-adopted IFRS
Cash Flow Statement (direct method, must reconcile)	IAS 7.	IAS 7

TOPIC	WHAT WE FOLLOWED	SOURCE
VAT 18% · mandatory EFD fiscal receipts	TRA / EFDMS rules.	tra.go.tz
VAT-withholding (3% goods / 6% services, from 1 Jul 2025) · 2% govt WHT	Finance Act 2025.	EY / RSM Tanzania tax guides
Payroll: PAYE (progressive), NSSF 20%, SDL 3.5%, WCF 0.5%	Tanzania employment-tax rules.	PwC Tanzania
Mobile money as the dominant payment method (M-Pesa, Mixx, Airtel, HaloPesa)	Captured as payment methods on receipts/payments.	TanzaniaInvest / ClickPesa
Fixed assets, depreciation, disposal	IAS 16 / IFRS for SMEs §17 (book), capital allowances tracked separately for tax.	Income Tax Act
Accounting standards regulator	NBAA; IFRS & IFRS for SMEs adopted in Tanzania.	nbaa.go.tz

The honest current status. Most money doors post correctly today. A few are still being wired to the one ledger (notably **POS sales & POS returns**). Those are marked “being wired” in the tables above so this guide stays truthful about what is and isn't live. The full door-by-door audit, with the exact double entry and status for each, lives in the project's `money.md` .

BMS — How Money Moves & How the Reports Are Built. Prepared for BJP Technologies Co. Ltd. Generated from the live system design (the one ledger in `core/financial_reports.php` and the money-flow audit in `money.md`). To save as PDF: open in any browser and **Print** → **Save as PDF**.